Composite Structures 261 (2021) 113320

Contents lists available at ScienceDirect

Composite Structures

journal homepage: www.elsevier.com/locate/compstruct

On the use of non traditional stacking to maximize critical buckling loads in flat composite panels

Avinkrishnan Vijayachandran [a], Anthony Waas [b,*]

[a] Graduate Student Research Assistant, Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI 48105, United States
[b] Richard A. Auhll Department Chair, Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI 48105, United States

ARTICLE INFO

Keywords:
Buckling
Unsymmetrical laminates
Optimization
DD laminates
Manufacturability
Eigenvalue problem

ABSTRACT

Traditional lay-ups in laminated composite structural parts have largely focused on using quasi-isotropic laminates made of permutations of $0^\circ, 90^\circ$ and $\pm 45^\circ$ laminae. These laminates are referred to as legacy quad laminates (LQL). In this study, three different lay-ups of symmetric laminates to compute critical buckling loads under uniaxial in-plane compressive loads is first investigated. An optimization problem is solved to determine the lay-ups that maximize the buckling performance. The resulting laminates are compared against a new class of laminates, referred to as double-double (DD), which have distinctive advantages for thickness tapering and therefore manufacturability. Both, symmetric and unsymmetric DD laminates are studied, while in the latter case, a thermal cure cycle followed by the application of in-plane loads is used to determine the structural performance in compression. To further verify the versatility of DD laminates, biaxial buckling loads are computed for quad, DD and optimized lay-ups. It is concluded that DD laminates, because of their manufacturability advantages, are promising candidates for several aerostructural applications.

1. Introduction

Lightweight, stiffness critical aerospace structures are usually thin gage, ranging in thickness from 8-plies (typical pre-preg tape based laminates are used here) to about 48 plies of fiber reinforced lamine, where a nominal lamina (ply) thickness is about 0.0075 in (0.2 mm). Buckling under various loading cases is usually a design driver, setting the skin thickness of wings, fuselages, wing boxes, rocket wall thicknesses etc. At the same time, an optimized design calls for non-uniform skin thickness, usually requiring taper. The taper is acheived by ply-drops in traditional manufacturing and may result in resin pockets that can act as stress raisers and regions of lowered local stiffness. Therefore, the quest to manufacture tapered and non-tapered high performing structures efficiently at minimum cost is a continuing challenge.

Many airframers and other mnaufacturers of lightweight aerostructures have risen to this challenge. Robotically manufactured, steered fiber panels and woven structures are two of the more recent advances that will undoubtedly find significant applications in the future. At the same time, stacking sequences that provide the best pathway for producing thickness tapering using pre-preg based laminates, while maximizing structural performance, is also a challenge that needs to be addressed.

Recently, a new class of laminates, referred to as double-double (DD) [1] has been studied for applications to aerospace structures because of the inherent advantages associated with manufacturability, especially thickness tapering and ease of automation. In these laminates, the stacking sequence is dictated by a lay-up that has the form, $(\theta_1/-\theta_1/\theta_2/-\theta_2)$, where the signs are interchangable but the angle pairs occur in that order. Stacks of these for appropriate values of θ_1 and θ_2 lead to laminates that have shown ease of manufacturability and thickness tapering. Therefore, a natural question that arises is their structural performance under in-plane loads.

In this paper, we consider the buckling performance of a laminated square plate that is simply supported on all sides and subjected to uniaxial and biaxial loads. An optimization study produces the lay-ups that maximize the uniaxial buckling loads. These lay-ups are contrasted against the traditional quad-lay-ups and associated DD laminates. For the DD class, both symmetric and unsymmetric lay-ups are studied. In the latter case, in addition to the response problem of a perfectly flat laminate, a thermal cycle is used to first manufacture the laminate, followed by the application of in-plane compressive

* Corresponding author.
E-mail addresses: avinav@umich.edu (A. Vijayachandran), awaas@umich.edu (A. Waas).

https://doi.org/10.1016/j.compstruct.2020.113320
Received 24 August 2020; Accepted 7 November 2020
Available online 19 November 2020

loads, in order to determine the magnitude of load that pushes the laminate to the post-buckled regime.

The paper is organized as follows. We first provide the notions of buckling loads and transaction loads (for unsymmetric laminates). Next we describe the optimization study and finally discuss the results for unsymmetric DD laminates. The paper closes with two biaxial loading cases and concluding remarks.

2. Buckling loads and transition loads

Fig. 1 describes the co-ordinates used and Fig. 2 illustrates the in-plane load and moment resultants in an infinitesimal element of a laminated composite plate. Using Classical Lamination Theory, a constitutive relationship between the force and moment resultants (N and M), and the strains and curvatures, (ε and χ, are established as in Eqn. (1) [2]

$$\begin{Bmatrix} N \\ M \end{Bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{Bmatrix} \varepsilon^0 \\ \chi \end{Bmatrix} \tag{1}$$

where,

$A-$ Extensionalstiffnessmatrix
$B-$ Coupling(extensional $-$ bending)stiffnessmatrix
$D-$ Bendingstiffnessmatrix

$$\begin{aligned} A_{ij} &= \sum_{k=1}^{n} \overline{Q}_{ij}^{k}(h_k - h_{k-1}) \\ B_{ij} &= \tfrac{1}{2}\sum_{k=1}^{n} \overline{Q}_{ij}^{k}(h_k^2 - h_{k-1}^2) \\ D_{ij} &= \tfrac{1}{3}\sum_{k=1}^{n} \overline{Q}_{ij}^{k}(h_k^3 - h_{k-1}^3) \end{aligned} \tag{2}$$

The geometrically nonlinear Von-Karman strains, ε and curvatures χ, in terms of the mid-plane displacements, $u_0(x,y), v_0(x,y)$ and $w(x,y)$, are,

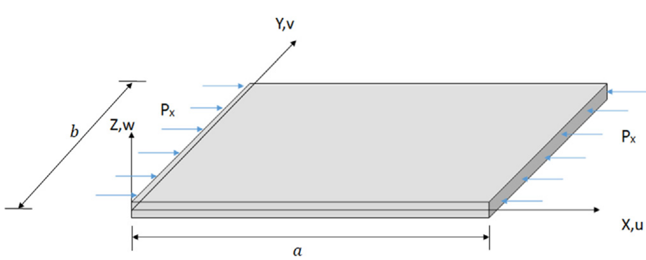


Fig. 1. Co-ordinates used with nomenclature for displacement components.

$$\begin{aligned} \varepsilon_x &= \tfrac{\partial u_0}{\partial x} - z\tfrac{\partial^2 w}{\partial x^2} + \tfrac{1}{2}\left(\tfrac{\partial w}{\partial x}\right)^2 \\ \varepsilon_y &= \tfrac{\partial v_0}{\partial y} - z\tfrac{\partial^2 w}{\partial y^2} + \tfrac{1}{2}\left(\tfrac{\partial w}{\partial y}\right)^2 \\ \varepsilon_{xy} &= \tfrac{\partial u_0}{\partial y} + \tfrac{\partial v_0}{\partial x} - 2z\tfrac{\partial^2 w}{\partial x \partial y} + \tfrac{\partial w}{\partial x}\tfrac{\partial w}{\partial y} \end{aligned} \tag{3}$$

$$\begin{Bmatrix} \chi_x \\ \chi_y \\ \chi_{xy} \end{Bmatrix} = \begin{Bmatrix} -\frac{\partial^2 w_0(x,y)}{\partial x^2} \\ -\frac{\partial^2 w_0(x,y)}{\partial y^2} \\ -2\frac{\partial^2 w_0(x,y)}{\partial x\ \partial y} \end{Bmatrix} \tag{4}$$

We consider an initially flat laminate that is subjected to only in-plane loads. The laminate equilibrium equations are then written in the current deformed configuration, where a general point on the mid-plane of the laminate can have non-zero axial displacements and out of plane displacements, respectively. Eqs. (5) and (6), represents the in-plane equations of equilibrium and Eq. (7) represents the out of plane equation for equilibrium, [2].

$$\frac{\partial N_x}{\partial x} + \frac{\partial N_{xy}}{\partial y} = 0$$

$$\frac{\partial N_{xy}}{\partial x} + \frac{\partial N_y}{\partial y} = 0$$

$$N_x\frac{\partial^2 w}{\partial x^2} + 2N_{xy}\frac{\partial^2 w}{\partial x \partial y} + N_y\frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = 0$$

2.1. Linearization using adjacent equilibrium method

The most general form of the laminated plate equations of equilibrium as described in Eqs. (5), (6) and (7) are fully coupled, due to the $[B_{ij}]$ matrix and nonlinear because of the terms related to the product of in-plane stress resultants and plate curvatures. Thus, in general, a stacked up laminate will deform out of plane (nonlinearly) due to the applied loading in the plane of the laminate, as long as the coupling matrix $[B_{ij}] \neq 0$. In general, a trivial solution, with $w = 0$ does not exist for all values of the applied in-plane loads that also satisfy the boundary conditions. Therefore, we now define the concept of a critical buckling load, based on a *linear* analysis. The procedure used to linearize the equilibrium equations is known as the *Adjacent Equilibrium* method [3,4] in stability analysis and is described below.

Consider the displacement field

$$\begin{aligned} u_0 &= u_0^i + \lambda u_0 \\ v_0 &= v_0^i + \lambda v_0 \\ w &= w^i + \lambda w \end{aligned} \tag{8}$$

where 'i' denotes pre-buckling displacements and λ is an infinitesimally small, dimensionless quantity. A critical load that causes an infinitesimal shift in the initial equilibrium position is sought. Thus, u_0, v_0 and w are the buckling displacements associated with the shift in the equilibrium position from the initial to the buckled.

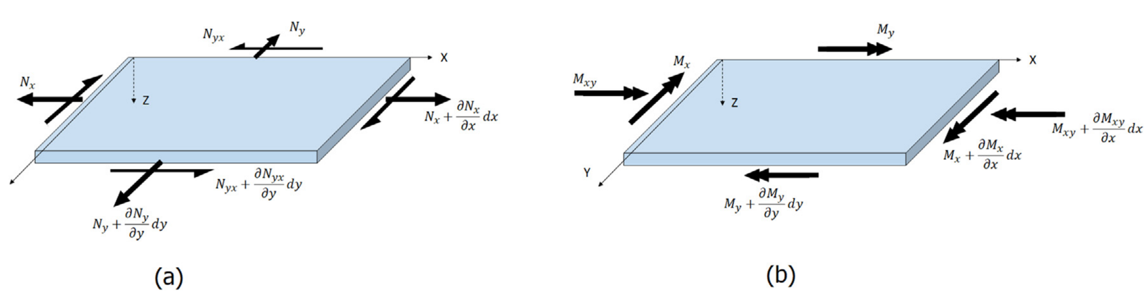


Fig. 2. (a) Force and (b) Moment resultants on an infinitesimal plate element.

Eq. (8) in conjunction with the plate constitutive description Eq. (1) is used to arrive at

$$\begin{aligned} N &= A\varepsilon^{0i} + B\chi^i + \lambda(A\varepsilon^0 + B\chi) = N^i + \lambda N \\ M &= B\varepsilon^{0i} + D\chi^i + \lambda(B\varepsilon^0 + D\chi) = M^i + \lambda M \end{aligned} \tag{9}$$

Using Eqs. (8) and (9) in Eq. (7) and collecting terms of like powers in λ and neglecting second order terms in λ we arrive at

$$\begin{aligned} &N_x^i\frac{\partial^2 w}{\partial x^2} + N_x\frac{\partial^2 w^i}{\partial x^2} + 2N_{xy}^i\frac{\partial^2 w}{\partial x\partial y} + 2N_{xy}\frac{\partial^2 w^i}{\partial x\partial y} + N_y^i\frac{\partial^2 w}{\partial y^2} + N_y\frac{\partial^2 w^i}{\partial y^2} \\ &\quad + \frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x\partial y} + \frac{\partial^2 M_y}{\partial y^2} \\ &\quad = 0 \end{aligned} \tag{10}$$

Since w^i, N_x^i, N_y^i and N_{xy}^i are obtained from the solution to the initial equilibrium position, Eq. (10) is linear. However, we still need to use the non-linear version of Eq. (7) to determine the initial configuration. At this stage, we make an assumption that linear theory be used to determine the solution corresponding to the initial equilibrium position. Since the initial configuration is determined from linear theory, the terms in Eq. (10) containing initial curvatures can be neglected. Thus, Eq. (10) becomes

$$N_x^i\frac{\partial^2 w}{\partial x^2} + 2N_{xy}^i\frac{\partial^2 w}{\partial x\partial y} + N_y^i\frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x\partial y} + \frac{\partial^2 M_y}{\partial y^2} = 0 \tag{11}$$

Then, Eq. (11) together with equations Eqs. (5) and (6) complete the governing equations for the stability (buckling) problem. The linear prebuckling problem, corresponding to the initial equilibrium configuration, is governed by the three pre-buckling equations, Eqs. (12), (13) and (14), as shown below.

$$\frac{\partial N_x^i}{\partial x} + \frac{\partial N_{xy}^i}{\partial y} = 0$$

$$\frac{\partial N_{xy}^i}{\partial x} + \frac{\partial N_y^i}{\partial y} = 0$$

$$\frac{\partial^2 M_x^i}{\partial x^2} + 2\frac{\partial^2 M_{xy}^i}{\partial x\partial y} + \frac{\partial^2 M_y^i}{\partial y^2} + N_x^i\frac{\partial^2 w^i}{\partial x^2} + 2N_{xy}^i\frac{\partial^2 w^i}{\partial x\partial y} + N_y^i\frac{\partial^2 w^i}{\partial y^2} = 0$$

Since the initial prebuckling state is assumed to be associated with negligible initial curvatures, Eq. (14) can be reduced to Eq. (14a)

$$\frac{\partial^2 M_x^i}{\partial x^2} + 2\frac{\partial^2 M_{xy}^i}{\partial x\partial y} + \frac{\partial^2 M_y^i}{\partial y^2} = 0 \tag{15}$$

In terms of displacements, the buckling equations, Eq. (5), Eq. (6) and (11) become,

$$\begin{aligned} &A_{11}\frac{\partial^2 u_0}{\partial x^2} + 2A_{16}\frac{\partial^2 u_0}{\partial x\partial y} + A_{66}\frac{\partial^2 u_0}{\partial y^2} + A_{16}\frac{\partial^2 v_0}{\partial x^2} + (A_{12}+A_{66})\frac{\partial^2 v_0}{\partial x\partial y} \\ &\quad + A_{26}\frac{\partial^2 v_0}{\partial y^2} - B_{11}\frac{\partial^3 w}{\partial x^3} - 3B_{16}\frac{\partial^3 w}{\partial x^2\partial y} - (B_{12}+2B_{66})\frac{\partial^3 w}{\partial x\partial y^2} - B_{26}\frac{\partial^3 w}{\partial y^3} = 0 \\ &A_{16}\frac{\partial^2 u_0}{\partial x^2} + (A_{12}+A_{66})\frac{\partial^2 u_0}{\partial x\partial y} + A_{26}\frac{\partial^2 u_0}{\partial y^2} + A_{66}\frac{\partial^2 v_0}{\partial x^2} + 2A_{26}\frac{\partial^2 v_0}{\partial x\partial y} \\ &\quad + A_{22}\frac{\partial^2 v_0}{\partial y^2} - B_{16}\frac{\partial^3 w}{\partial x^3} - (B_{12}+2B_{66})\frac{\partial^3 w}{\partial x^2\partial y} - 3B_{26}\frac{\partial^3 w}{\partial x\partial y^2} - B_{22}\frac{\partial^3 w}{\partial y^3} = 0 \\ &D_{11}\frac{\partial^4 w}{\partial x^4} + 4D_{16}\frac{\partial^4 w}{\partial x^3\partial y} + 2(D_{12}+2D_{66})\frac{\partial^4 w}{\partial x^2\partial y^2} + 4D_{26}\frac{\partial^4 w}{\partial x\partial y^3} + D_{22}\frac{\partial^4 w}{\partial y^4} \\ &\quad - B_{11}\frac{\partial^3 u_0}{\partial x^3} - 3B_{16}\frac{\partial^3 u_0}{\partial x^2\partial y} - (B_{12}+2B_{66})\frac{\partial^3 u_0}{\partial x\partial y^2} - B_{26}\frac{\partial^3 u_0}{\partial y^3} - B_{16}\frac{\partial^3 v_0}{\partial x^3} \\ &\quad - (B_{12}+2B_{66})\frac{\partial^3 v_0}{\partial x^2\partial y} - 3B_{26}\frac{\partial^3 v_0}{\partial x\partial y^2} - B_{22}\frac{\partial^3 v_0}{\partial y^3} = N_x^i\frac{\partial^2 w}{\partial x^2} + 2N_{xy}^i\frac{\partial^2 w}{\partial x\partial y} \\ &\quad + N_y^i\frac{\partial^2 w}{\partial y^2} \end{aligned}$$

2.2. Boundary conditions

For the purpose of this paper, we will consider a laminate that is subjected to simply supported boundary conditions. With the assumed linearization, as shown in Eqn. (8), the boundary conditions for the pre-buckling and buckling problem are as follows.

For pre-buckling, At $x = 0$ and $x = a$

$$\begin{aligned} N_x^i &= -P_x \\ w^i &= 0 \\ N_{xy}^i &= M_x^i = 0 \end{aligned} \tag{19}$$

At $y = 0$ and $y = b$

$$\begin{aligned} v_0^i &= w^i = 0 \\ N_{xy}^i &= M_y^i = 0 \end{aligned} \tag{20}$$

and for buckling, the homogenous boundary conditions are, At $x = 0$ and $x = a$

$$\begin{aligned} N_x &= 0 \\ w &= 0 \\ N_{xy} &= M_x = 0 \end{aligned} \tag{21}$$

At $y = 0$ and $y = b$

$$\begin{aligned} v_0 &= w = 0 \\ N_{xy} &= M_y = 0 \end{aligned} \tag{22}$$

We next consider the solution of the pre-buckling and buckling equations for symmetric and unsymmetric laminates. We will see that a trivial solution that satisfies the pre-buckling equations and the pre-buckling boundary conditions does not exist, in general, for an unsymmetric laminate. Special lay-ups, such as angle ply laminates with specific sets of boundary conditions may yield a trivial solution. As discussed in [5,6], a generally laminated plate with a non-zero coupling matrix starts to deform out-of-plane as soon as in-plane edge loading is applied. Thus, the concept of a buckling (bifurcation) load, therefore, only exists for symmetric laminates and for some special cases of unsymmetric laminates, with the assumption of negligible initial curvatures. Consequently, we solve a geometrically non-linear response problem for the unsymmetric laminate from which we extract the transition load. The transition load is the load at which a significant change in the plate axial stiffness is seen to take place. This load can be extracted from a plot of the external edge load vs. displacement response plot, as shown later in this paper. Furthermore, we observe that for external loads below the transition load, the out-of-plane displacements and plate curvatures remain small. Consequently, the transition load is also the load beyond which the plate curvatures become large.

2.3. Symmetric laminates

In cases I, II, III to be studied later, it is to be noted that the lay-up is symmetric. Thus invoking the definitions in Eq. (2), one could see that the coupling matrix $[B_{ij}] = 0$. Thus, the terms containing $[B_{ij}]$ in Eq. (16), Eqs. (17) and (18) vanish, thereby uncoupling the in-plane equations from the out-of plane equation.

Upon inspection, $N_x^i = -P_x, N_y^i = 0, N_{xy}^i = 0$, in the initially flat pre-buckled state, automatically satisfies the uncoupled in-plane Eqs. (5) and (6). Thus, with this initial solution, Eq. (18) is now modified as

$$\begin{aligned} &D_{11}\frac{\partial^4 w}{\partial x^4} + 4D_{16}\frac{\partial^4 w}{\partial x^3\partial y} + 2(D_{12}+2D_{66})\frac{\partial^4 w}{\partial x^2\partial y^2} + 4D_{26}\frac{\partial^4 w}{\partial x\partial y^3} + D_{22} \\ &\quad \times \frac{\partial^4 w}{\partial y^4} \\ &\quad = N_x^i\frac{\partial^2 w}{\partial x^2} \end{aligned} \tag{23}$$

Upon inspection $w = 0$ is a trivial solution, but we seek the existence of a non-trivial solution that will satisfy this equation and the associated homogeneous boundary conditions, Eqs. (21) and s(22). Since an analytical closed-form solution for Eq. (23) is not available, numerical methods like the Rayleigh–Ritz method or Finite Element Analysis (FEA) [2] are sought to obtain the solution. In this work, an FEA formulation using the commercial software Abaqus is adopted.

2.4. Unsymmetric laminates

In Case IV, an unsymmetric laminate of the type DD is analyzed. Two problems for this case are considered. First, a fully non linear response problem is analyzed for an initially flat unsymmetric laminate. The expected behavior would dictate a bilinear response, as stated earlier. The load at the intersection of the two stable equilibrium paths is considered as the transition load for this case. Second, to examine the knock down of the transition load due to initial curvatures that could be caused during the manufacturing, a thermal analysis is first conducted to compute the initial deformation. Next, in-plane edge loading is applied in conjunction with a geometrically nonlinear analysis to capture the reduction in the transition load.

3. Problem statement

A flat, square panel, of size 20 in $\times$ 20in, that is simply supported on all the edges (S4) with an applied in-plane uniaxial compressive load as shown in Fig. 3, is considered. An 8ply, symmetric configuration is considered as the layup. In general this is represented as $[\theta_1/\theta_2/\theta_3/\theta_4]_s$ The material properties of T800/3900S are used, and are detailed in Table 1.

The optimization problem seeks the maximization of the critical buckling loads for the three symmetric layup cases, $[\theta_1/\theta_2]_{2s}$,$[\theta_1/-\theta_1/\theta_2/-\theta_2]_s$ and a general $[\theta_1/\theta_2/\theta_3/\theta_4]_s$. The optimization problem seeks the values of θ_i, where, $i = 1, 2$, for the first two cases, and $i = 1-4$, for the last case, that would maximize the critical buckling load. The improvement in performance is then compared to the critical buckling load of a legacy quad laminate (LQL) that is commonly used for aerospace applications. It is to be noted that since the number of layers and material remains the same, the mass of all these layups and LQL will remain the same.

Further, for case II laminates, generally referred to as DD or Double-Double [1], a corresponding unsymmetric lay-up $[\theta_1/-\theta_1/\theta_2/-\theta_2]_2$ is also considered in case IV. No optimization is considered for case IV, but the effect of unsymmetric lay-up in the critical transition load is analyzed. First, a response problem of an initially flat panel of this unsymmetric lay-up is discussed, comparing the reduction in transition load to case II. Secondly, since this unsymmetric lay-up is expected to not remain flat after manufacturing, a response problem is considered by first invoking a curing process from $350^\circ F$ to $70^\circ F$, and then obtaining the load displacement response when a compressive in-plane load is applied. After computing the buckling and transition loads for the above four cases, biaxial buckling under 1:1 and 2:1 bi-axial in-plane loads are investigated.

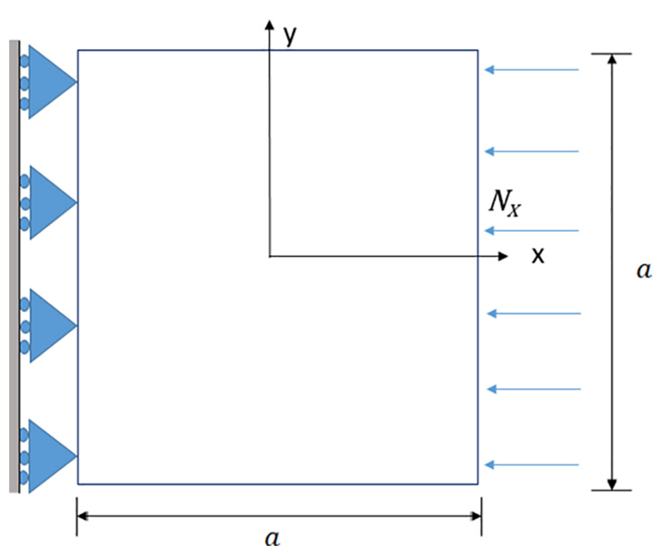


Fig. 3. Loading and boundary conditions.

Table 1
Material properties of T800/3900S.

Property	Value	Unit
E_{11}	21.5	Msi
E_{22}	1.23	Msi
ν_{12}	0.329	
G_{12}	0.571	Msi
α_1	-0.1	$\mu m/m\,^\circ C$
α_2	31.6	$\mu m/m\,^\circ C$

4. Finite element model

A commercial FEA package, Abaqus was used to perform the recursive finite element calculations during the optimization process. A python code was set up in Abaqus that would interact with the optimizer to provide the values of objective function in every iteration. Composite shell elements were used with a pixelated mesh of 100 $\times$ 100 grid containing 10201 nodes, with each node having three translational and two rotational degrees of freedom.

4.1. Symmetric case

The developement of the weak form of Eq. (23) is provided in [2]. A flexural stiffness matrix $[K]$ is obtained from the weakform corresponding to the terms on the left side of Eq. (23). A geometric stiffness matrix $[G]$, due to a unit distributed load is obtained from the weak form corresponding to the terms on the right side, thus resulting in an eigenvalue problem as shown in Eq. (24).

$$[K - \lambda_i G]\Phi_i = 0 \tag{24}$$

where, λ_i is the ith eigenvalue and Φ_i is the corresponding eigenmode. The critical buckling load corresponds to the lowest eigenvalue.

4.2. Unsymmetric case

Following the weak form formulations for the plate buckling equations Eqs. (16), (17) and (18), an eigenvalue problem can be set up for this case, though as mentioned earlier, a fully nonlinear response analysis is conducted to find the effect of unsymmetric layup on the transition load. In setting up the eigenvalue problem and proceeding, it turns out that a trivial solution, $w^i = 0$ does not exist for any value of the in-plane external loads. Thus, no eigenvalues exist for the unsymmetric case considered here, and therefore a buckling (bifurcation) load does not exist. However, by performing a geometrically nonlinear response analysis, a transition load can be identified. It is noted that one can seek an eigenvalue for the unsymmetric case considered in this study, using the commercial code Abaqus, but this value does not correspond to any meaningful aspect of the unsymmetric plate response.

We first study the load–displacement response to compute the transition load. A bilinear behavior is expected for the response and a transition load is calculated approximately at the intersection of the slopes of the two stable paths. Next, to capture the effect of manufacturing, at first a curing simulation is carried out, subjecting the laminate to a temperature change from 350°F to 70°F. This would provide the initial geometry of the panel after the cure cycle. Next, a response analysis is conducted in which in-plane compressive load is applied to seek the response of the plate and a transition load is identified.

5. Optimization problem

An optimization problem is set up to maximize the buckling load for three different lay-ups- $[\theta_1/\theta_2]_{2s}$,$[\theta_1/-\theta_1/\theta_2/-\theta_2]_s$ and $[\theta_1/\theta_2/\theta_3/\theta_4]_s$, where, $i = 1, 2$, for the first two cases, and $i = 1-4$, for the third case, respectively. The angles represent the straight fiber angle of each layer. Hence, these θ_i's become the optimization variables. The bounds on the fiber angles are $-90° \leqslant \theta_i \leqslant 90°$. The mathematical representation of the optimization problem is as seen in Eq. (25)

$$\begin{array}{ll} \min\limits_{X} & (-N_{CR}) \\ \text{subjectto} & -90° \leqslant \theta_i \leqslant 90° \end{array} \tag{25}$$

where N_{CR} is the critical buckling load, θ_i, ($i = 1, 2$ or $i = 1-4$) are the fiber angles A general optimization work flow is shown in Fig. 4.

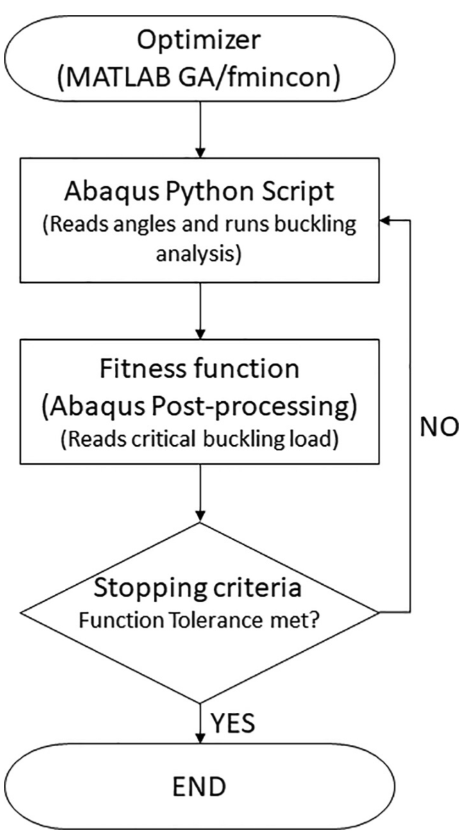


Fig. 4. Workflow for optimization set-up.

6. Results

The three cases studied are described as below. Table 2 shows the comparison of the performance of these results with Legacy Quad $[0/90/\pm45]_s$ laminates (see Fig. 5).

Table 2
Comparison of optimal solutions for case I, II, III.

Case	Layup	N_{cr} - lb/in (N/mm)
Unidirectional	$[0]_8$	11.88 (2.08)
Legacy Quad	$[0/90/\pm45]_s$	9.79 (1.71)
Case I	$[\pm27]_{2s}$	13.23 (2.32)
Case II	$[\pm41/\pm2.5]_s$	15.4 (2.69)
Case III	$[41/-42/-6/0]_s$	15.46 (2.71)

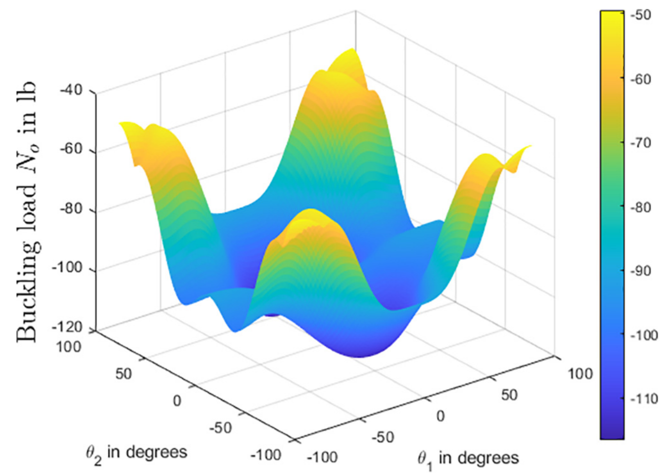


Fig. 5. Response surface obtained for critical buckling loads.

6.1. Case I: $[\theta_1/\theta_2]_{2s}$

Here an 8 ply laminate is considered with the layup $[\theta_1/\theta_2]_{2s}$. A coarse sweep of the design space was conducted to determine the convexity of the objective function. fmincon, which is Sequential Quadratic Programming (SQP) function in-built in the MATLAB ®optimization toolbox was used to conduct optimization starting at various initial points near the cusps of the objective function in the design space. Since SQP is a gradient based solver, it converges to the closest local minimum. In such a layup, multiple minimum were observed. It is interesting to note that, while intuition suggests an 8-ply unidirectional laminate $[0]_8$ might provide the best buckling load, it is only one of the local minima of the objective function.Global minima was observed at $[27/-27]_{2s}$ with buckling load 13.23 lb/in (2.32 N/mm) compared to 11.88 lb/in (2.08 N/mm) and 9.79 lb/in (1.71 N/mm) of unidirectional and LQL $[0/90/\pm45]_s$ respectively. Thus these designs offer increased buckling loads of up to 35% com-

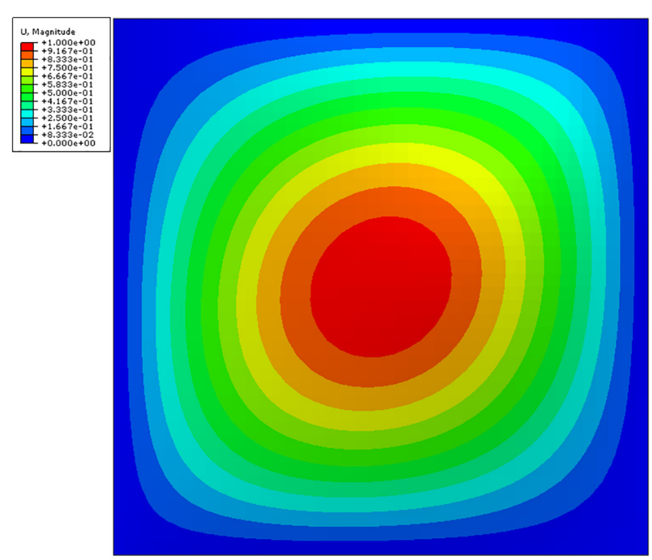


Fig. 6. Critical mode shape for $[27/-27]_{2s}$.

pared to LQL. The corresponding mode shapes for $[27/-27]_{2s}$, is shown in Fig. 6.

6.2. Case II: $[\theta_1/-\theta_1/\theta_2/-\theta_2]_s$

In this case, an 8-layer laminate of type DD (Double-Double)[1] is considered. A symmetric layup is considered, as an unsymmetric layup involves a fully populated $[B]$ matrix, and this is relegated to case IV. Similar to as described for Case I, SQP functionality of MATLAB ®was used to perform optimization which converged to a global optimum of $[+41/-41/+2.5/-2.5]_s$. The critical buckling load for this design is 15.4 lb/in (2.69 N/mm) which is 58% higher than LQL $[0/90/\pm45]_s$ which has a critical buckling load of 9.79 lb/in(1.71 N/mm).The buckling mode is shown in Fig. 7.

6.3. Case III: $[\theta_1/\theta_2/\theta_3/\theta_4]_s$

In this case, a most general 8-ply, symmetric laminate is considered with each lamina having a discrete fiber angle. While in case I and case II, plotting a response surface was straight forward in an $\mathbb{R}^2$ design space, it was easy to identify points close to a local minima and assign them as initial guess for the SQP application, which converges to the optimum quickly. In this case, we have a $\mathbb{R}^4$ design space and it might be easier to do a direct optimization without a variable sweep. Taking advantage of the less computational time for the FE model, a global optimization technique is used. Genetic Algorithm (GA) function in MATLAB ®global optimization toolbox was used for the same. The global optimum obtained was $[41/-42/-6/0]_s$, with a critical buckling load of 15.46 lb/in (2.71 N/mm), only higher than Case II by 0.4%. The mode shape is shown in Fig. 8.

Table 2 describes the optimal solutions of all the cases in comparison to Legacy Quad and Unidirectional Layups.

6.4. Case IV: response of $[\theta_1/-\theta_1/\theta_2/-\theta_2]_2$

In the case of unsymmetric laminate the lay-up considered uses the optimal angles from case II as $[+41/-41/+2.5/-2.5]_2$. In the first analysis, a non linear response of the initially flat panel is studied. It shows a clear bi-linear behavior in the axial load v/s axial deformation $(P-\Delta)$ plot. This response is then used to calculate the transition load at the intersection of the slopes of these stable paths as shown in Fig. 9. The transition load is computed as 225 lb (1kN) and an equivalent distributed load of 11.25 lb/in (1.97 N/mm), thus causing a 27% reduc-

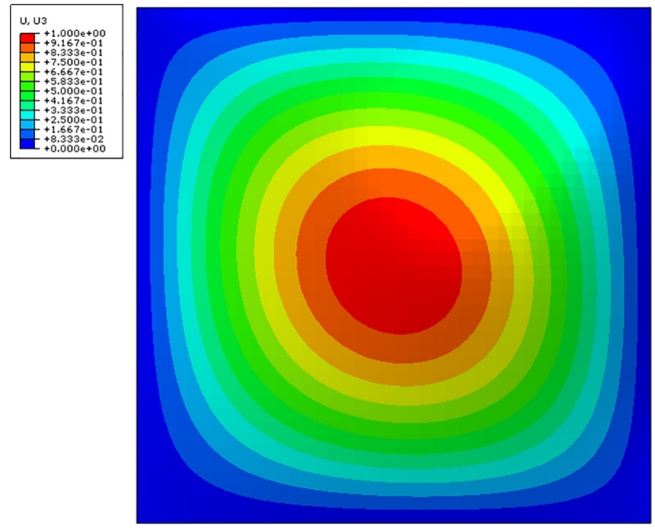


Fig. 8. Critical mode shape for $[41/-42/-6/0]_s$.

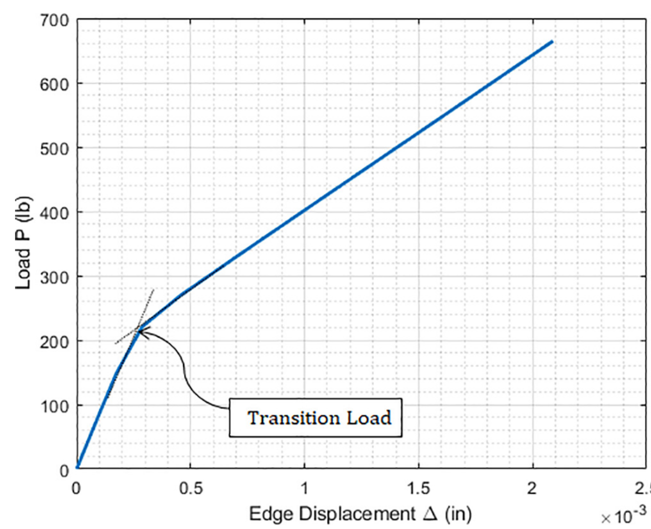


Fig. 9. $P-\Delta$ response of an initially flat $[+41/-41/+2.5/-2.5]_2$ panel.

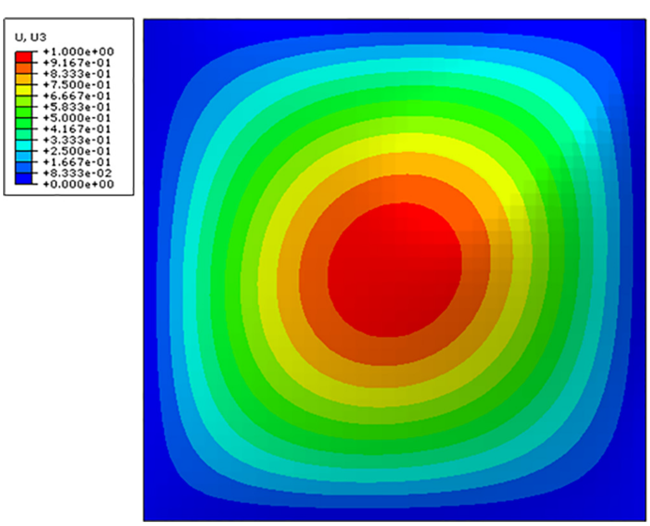


Fig. 7. Critical mode shape for $[+41/-41/+2.5/-2.5]_s$.

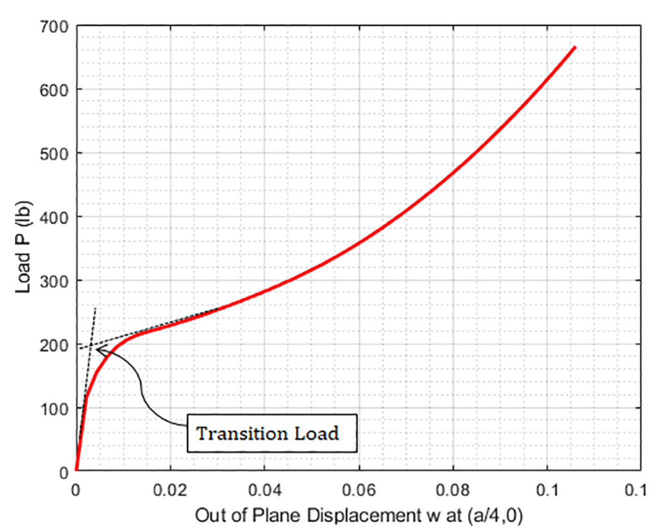


Fig. 10. $P-w$ out of plane displacement w at $x = a/4$, $y = 0$ for an initially flat $[+41/-41/+2.5/-2.5]_2$ panel.

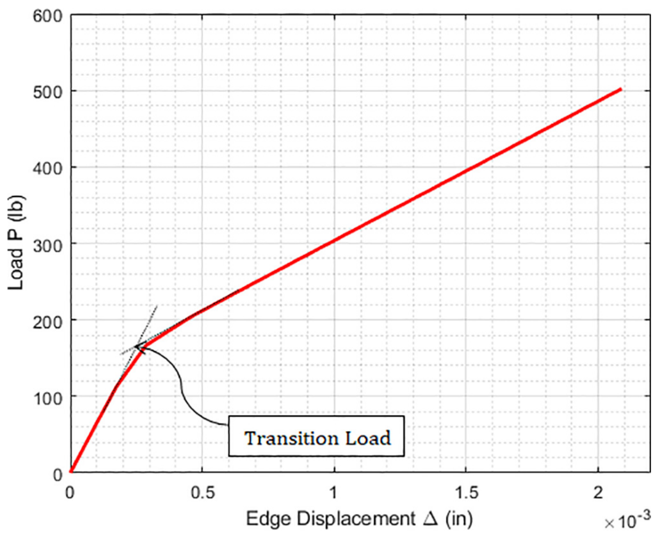


Fig. 11. $P-\Delta$ response for the post curing $[+41/-41/+2.5/-2.5]_2$ panel including identification of the transition load.

tion in the load compared to case II. A plot of load (P in lb) as a function of out of plane displacement (w in) at the location x = a/4, y = 0 is shown in Fig. 10. This plot also points to the transition in response at approximately 200 lb.

Next, a two step fully non linear analysis is conducted, where the first step is a thermal analysis to capture the effect of curing during manufacturing. The flat plate is first subjected to a thermal cycle from $350^{\circ}F$ to $70^{\circ}F$. Then a non-linear response analysis is conducted to obtain the transition load.

To obtain the transition load, the load–deflection curve is observed as shown in Fig. 11. An approximatley bi-linear response is observed. Slopes are drawn to match the stiffnesses before and after the point of transition. The intersection of these two slopes is the transition load and is calculated as 170 lb. This converts to an equivalent distributed load of 8.5 lb/in (1.49 N/mm), which is a 45% reduction when compared to the critical buckling load of the corresponding symmetric layup in case II. It is also lower than the transition load of the perfectly flat unsymmetric DD laminate response suggesting that the initial manufacturing can influence the axial load carrying capacity and hence the transition load. As the laminate becomes thicker, it is expected that the differences between the symmetric and unsymmetric laminate responses will diminish [5].

Therefore, a similar study was conducted for a 48 ply laminate. The transition loads of an initially flat $[+41/-41/+2.5/-2.5]_{12}$ and a cured panel of the same unsymmetric layup was compared to the buckling load of a $[+41/-41/+2.5/-2.5]_{6S}$ panel, also of 48 plies. The buckling load of the symmetric panel was computed as 2960 lb/in (518 N/mm). Transition loads for an initially flat unsymmetric panel was computed as 2800 lb/in (490 N/mm) and the cured panel was computed as 2750 lb/in (480 N/mm). The corresponding reduction in transition loads compared to the buckling load of the symmetric panel was 5% and7% respectively.

6.5. Response under biaxial in-plane loads

The three cases of non traditional lay-ups, corresponding to cases I-III have produced significant improvement of the critical buckling load under in-plane uniaxial loading. Among them case II, identified as DD, particularly stands out due to its easiness in manufacturing albeit providing one of the highest uniaxial buckling performance. It is imperative, therefore, that we discuss the performance of these designs under the effect of bi-axial in-plane loads. Two loading cases are considered, 1:1 biaxial loading and 2:1 biaxial loading as seen in Fig. 12.

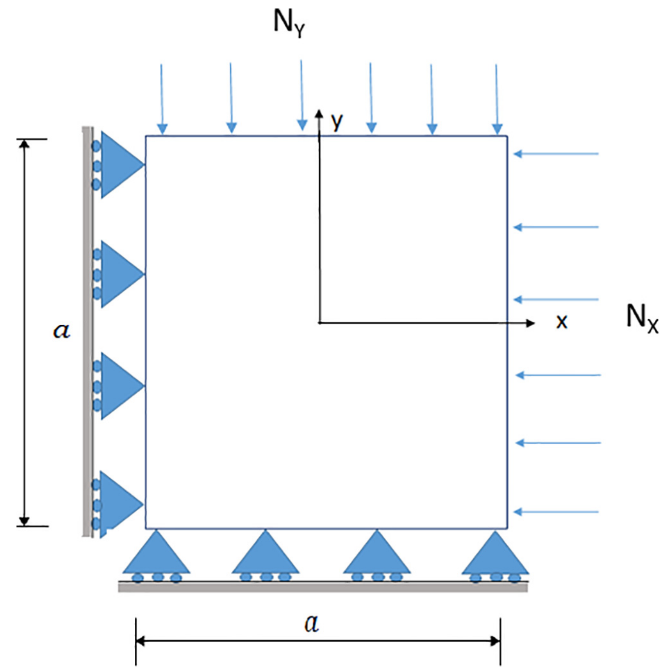


Fig. 12. Case V: Bi-axial loading.

6.5.1. 1:1 Biaxial loading

A 1:1 biaxial in-plane loading state, *viz., edge loads $N_y = N_x$, is considered. For the LQL lay-up $[0/90/\pm45]_s$, the critical buckling load in this case is 6.41 lb/in (1.13 N/mm). Each of the designs in cases I-III provide improved buckling performance. In case I, $[\pm27]_{2s}$ has a critical biaxial buckling load at 8.26 lb/in (1.45 N/mm), providing a corresponding 29 % increase compared to LQL. In cases II nd III, the lay-ups $[\pm45/\pm2.5]_s$ and $[41/-42/-6/0]_s$, provides a buckling load of 9.15 lb/in (1.60 N/mm) and 9.23 lb/in (1.62 N/mm) respectively with a corresponding improvement in buckling performance by 43 % and 44%. These results are tabulated in Table 3.*

6.5.2. 2:1 Biaxial loading

Similar to the above section, an analysis is conducted to investigate the buckling performance of the optimal solutions under a 2:1 biaxial in-plane loading state, *viz., edge loads $N_y = 2N_x$. For the LQL lay-up $[0/90/\pm45]_s$ considered, the critical buckling load in this case is 4.28 lb/in (0.75 N/mm). Similar to the 1:1 biaxial loading, each of the optimal designs in cases I-III provide improved buckling performance. In case I, $[\pm27]_{2s}$ has a critical biaxial buckling load at 5.51 lb/in (0.96 N/mm), providing a corresponding 29 % increase compared to LQL. In case II nd III, the lay-ups $[\pm45/\pm2.5]_s$ and $[41/-42/-6/0]_s$, provides a buckling load of 6.10 lb/in (1.07 N/mm) and 6.15 lb/in (1.08 N/mm) respectively with a corresponding improvement in buckling performance by 43 % and 44%. These results are tabulated in Table Table 4.*

Table 3
Comparison of 1:1 biaxial performance of optimal solutions.

Case	Layup	$1:1N_{cr}$ - lb/in (N/mm)
Legacy Quad	$[0/90/\pm45]_s$	6.41 lb/in (1.13 N/mm)
Case I	$[\pm27]_{2s}$	8.26 lb/in (1.45 N/mm)
Case II	$[\pm41/\pm2.5]_s$	9.15 lb/in (1.60 N/mm)
Case III	$[41/-42/-6/0]_s$	9.23 lb/in (1.62 N/mm)

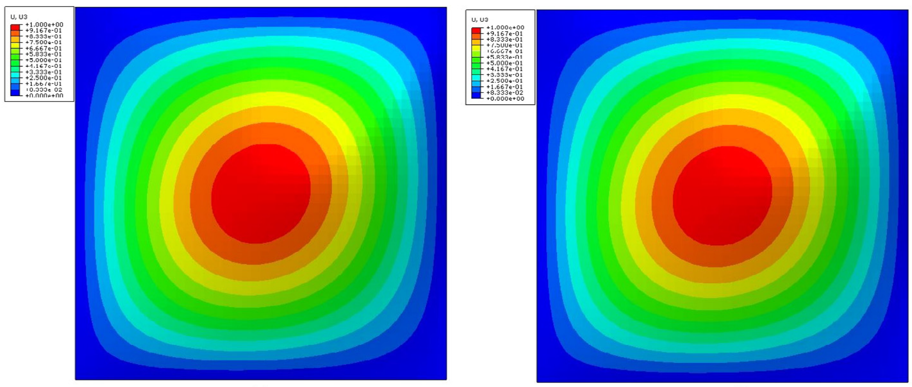
Table 4
Comparison of 2:1 biaxial performance of optimal solutions.

Case	Layup	$2:1N_{cr}$ - lb/in (N/mm)
Legacy Quad	$[0/90/\pm45]_s$	4.28 lb/in (0.75 N/mm)
Case I	$[\pm27]_{2s}$	5.51 lb/in (0.96 N/mm)
Case II	$[\pm41/\pm2.5]_s$	6.10 lb/in (1.07 N/mm)
Case III	$[41/-42/-6/0]_s$	6.15 lb/in (1.08 N/mm)

Figs. 13–15 shows the critical buckling mode shapes for each of the above cases for both $1:1$ and $2:1$ in-plane biaxial compressive loads.

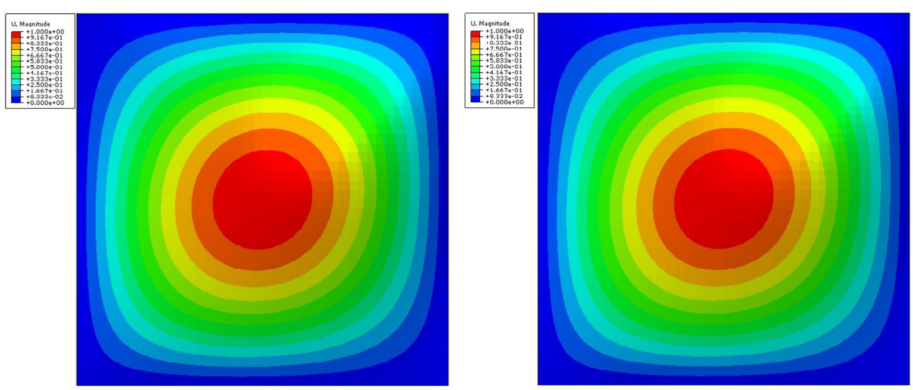
7. Conclusions

The buckling of a square flat laminated composite plate has been addressed with respect to laminate stacking. At first, three symmetric 8-ply laminates are considered of layups $[\theta_1/\theta_2]_{2s}$,$[\theta_1/-\theta_1/\theta_2/-\theta_2]_s$



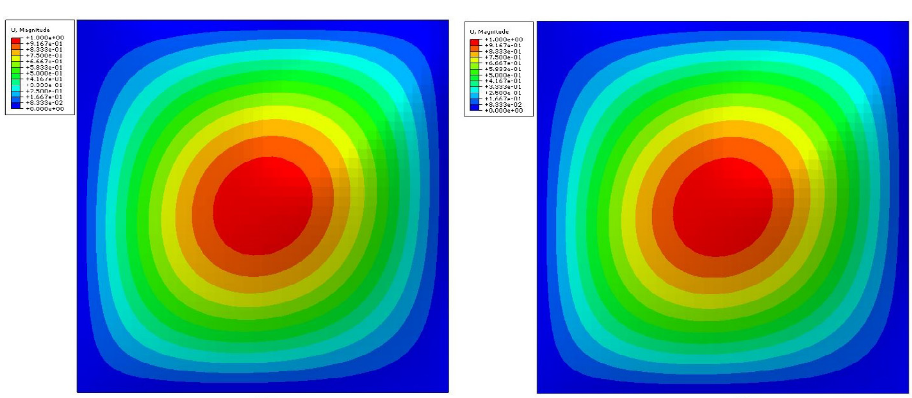
(a) (b)

Fig. 14. Critical Mode shape for $[\pm41/\pm2.5]_s$ (a) 1:1 Loading (b) 2:1 Loading.



(a) (b)

Fig. 15. Critical Mode shape for $[41/-42/-6/0]_s$ (a) 1:1 Loading (b) 2:1 Loading.



(a) (b)

Fig. 13. Critical Mode shape for $[\pm27]_{2s}$ (a) 1:1 Loading (b) 2:1 Loading.

and $[\theta_1/\theta_2/\theta_3/\theta_4]_s$. The optimal values of fiber angles θ_i are sought for each of the cases, and the results compared with Legacy Quad laminates ($[0/90/\pm45]_s$ that are Quasi-isotropic and very commonly used in aerospace applications, and a unidirectional $[0]_8$ laminate. Each of the three cases shows improved buckling loads compared to the benchmarks. Optimal Layups in Case I $[\pm27]_{2S}, [9/-19]_{2s}$ and $[-19/9]_{2s}$ shows up to 35% increase compared to legacy quads,where as Case II and Case III layups, $[\pm41/\pm2.5]_s$ and $[41/-42/-6/0]_s$ respectively, shows up to 58% increase in buckling loads. It is to be noted that since the geometry, material and thickness remains the same, the total mass of the panel remains the same in all the three cases, thus showing significant improvement in buckling load compared to legacy quad laminates.

In case of unsymmetric laminates $[\theta_1/-\theta_1/\theta_2/-\theta_2]_2$, curing induced initial out-of-flatness is first modeled and then a response analysis is conducted to obtain the transition load. A 48.5% reduction in the transition load was observed as compared to the symmetric layup $[\theta_1/-\theta_1/\theta_2/-\theta_2]_s$ of the same angles as in case II. However, as the thickness of the laminate increases, it was found that the effect of coupling was negligible and the initial our of flatness due to manufacturing had a negligible effect of the transition (buckling) load. In order to study the performance under biaxial in-plane compressive loads, 1:1 and 2:1 loading scenarios were examined. While all three designs in case I provided improved performance, only one of the case I designs showed a significant increase of 29% in critical buckling load in both loading cases. Furthermore, cases II and III designs showed up to 44% improvement in critical buckling loads in both bi-axial loading cases, suggesting that symmetric DD laminates, because of their performance and ease of manufacturability, are desirable candidates for thin-gage aerostructural applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grareful to Prof. Steve Tsai, Stanford University, Aeronautics Department, for encouragement, support and introducing DD laminates to the authors. The recent Stanford University workshop in DD laminates was most beneficial to the first author who is deeply grateful for the financial support to attend the workshop.

References

[1] Shrivastava S, Sharma N, Tsai S, Mohite PM. D and DD-drop layup optimization of aircraft wing panels under multi-load case design environment. Compos Struct; May 2020..

[2] Reddy JN. Mechanics of laminated composite plates and shells: theory and analysis. CRC Press, Taylor & Francis Group; 2003.

[3] Brush AO, Almroth BO. Buckling of bars, plates and shells. McGraw-Hill Book Company; 1975.

[4] Whitney JM. Structural analysis of laminated anisotropic plates. CRC Press, Taylor & Francis Group; 1987.

[5] Jones RM. Buckling and vibration of unsymmetrically laminated cross-ply rectangular plates. AIAA J 1973.

[6] Qatu MS, Leissa AW. Buckling or transverse deflections of unsymmetrically laminated plates subjected to in-plane loads. AIAA J; January 1973..